UNIVERSIDAD DE SAN CARLOS DE GUATEMALA

FACULTAD DE INGENIERIA

INGENIERIA EN CIENCIAS Y SISTEMAS

LABORATORIO DE INTRODUCCION A LA PROGRAMACIÓN Y COMPUTACIÓN 1

TAREA 8

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Sección A

GUATEMALA, 27 DE MARZO DE 2026

Capturas del Código

ConversorView.java:

```java
import javax.swing.*;
import java.awt.*;

Edit | Explain | Test | Document | Fix
public class ConversorView extends JFrame {

    private JTextField txtCelsius;
    private JButton btnFahrenheit;
    private JButton btnKelvin;
    private JLabel lblResultado;

    Edit | Explain | Test | Document | Fix
    public ConversorView() {
        setTitle("Conversor de Temperaturas");
        setSize( width: 450,  height: 300);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setResizable(false);

        // --- PALETA DE VERDES ---
        Color celesteFondo = new Color( r: 200,  g: 225,  b: 255);
        Color verdeOscuro = new Color( r: 34,  g: 85,  b: 34);
        Color verdeBorde = new Color( r: 150,  g: 190,  b: 150);
        // ------------------------

        JPanel panelPrincipal = new JPanel(new GridBagLayout());
        panelPrincipal.setBackground(celesteFondo);
        panelPrincipal.setBorder(BorderFactory.createEmptyBorder( top: 20,  left: 30,  bottom: 20,  right: 30));
```

```java
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets( top: 8,  left: 8,  bottom: 8,  right: 8);
gbc.fill = GridBagConstraints.HORIZONTAL;

JLabel lblTitulo = new JLabel( text: "Conversor de Temperaturas", SwingConstants.CENTER);
lblTitulo.setFont(new Font( name: "Segoe UI", Font.BOLD,  size: 18));
lblTitulo.setForeground(verdeOscuro);
gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
panelPrincipal.add(lblTitulo, gbc);

JLabel lblInstruccion = new JLabel( text: "Ingrese temperatura en grados Celsius:");
lblInstruccion.setFont(new Font( name: "Segoe UI", Font.PLAIN,  size: 12));
lblInstruccion.setForeground(new Color( r: 60,  g: 60,  b: 60));
gbc.gridy = 1; gbc.gridwidth = 2;
panelPrincipal.add(lblInstruccion, gbc);

txtCelsius = new JTextField();
txtCelsius.setFont(new Font( name: "Segoe UI", Font.PLAIN,  size: 14));
txtCelsius.setHorizontalAlignment(JTextField.CENTER);
txtCelsius.setBorder(BorderFactory.createCompoundBorder(
        BorderFactory.createLineBorder(verdeBorde,  thickness: 1),
        BorderFactory.createEmptyBorder( top: 6,  left: 8,  bottom: 6,  right: 8)
));
gbc.gridy = 2; gbc.gridwidth = 2;
panelPrincipal.add(txtCelsius, gbc);
```

```java
        btnFahrenheit = new JButton( text: "Convertir a Fahrenheit (°F)");
        btnFahrenheit.setFont(new Font( name: "Segoe UI", Font.BOLD,  size: 12));

        btnFahrenheit.setBackground(new Color( r: 46,  g: 139,  b: 87));
        btnFahrenheit.setForeground(Color.WHITE);
        btnFahrenheit.setOpaque(true);
        btnFahrenheit.setContentAreaFilled(true);
        btnFahrenheit.setBorderPainted(false);

        btnFahrenheit.setFocusPainted(false);
        btnFahrenheit.setCursor(new Cursor(Cursor.HAND_CURSOR));
        btnFahrenheit.setBorder(BorderFactory.createEmptyBorder( top: 10,  left: 12,  bottom: 10,  right: 12));
        gbc.gridy = 3; gbc.gridwidth = 1; gbc.gridx = 0;
        panelPrincipal.add(btnFahrenheit, gbc);

        btnKelvin = new JButton( text: "Convertir a Kelvin (K)");
        btnKelvin.setFont(new Font( name: "Segoe UI", Font.BOLD,  size: 12));

        btnKelvin.setBackground(new Color( r: 34,  g: 139,  b: 34));
        btnKelvin.setForeground(Color.WHITE);
        btnKelvin.setOpaque(true);
        btnKelvin.setContentAreaFilled(true);
        btnKelvin.setBorderPainted(false);
```

```java
        btnKelvin.setFocusPainted(false);
        btnKelvin.setCursor(new Cursor(Cursor.HAND_CURSOR));
        btnKelvin.setBorder(BorderFactory.createEmptyBorder( top: 10,  left: 12,  bottom: 10,  right: 12));
        gbc.gridx = 1;
        panelPrincipal.add(btnKelvin, gbc);

        lblResultado = new JLabel( text: "Resultado: ", SwingConstants.CENTER);
        lblResultado.setFont(new Font( name: "Segoe UI", Font.BOLD,  size: 14));
        lblResultado.setForeground(verdeOscuro);
        lblResultado.setBorder(BorderFactory.createCompoundBorder(
                BorderFactory.createLineBorder(verdeBorde,  thickness: 1),
                BorderFactory.createEmptyBorder( top: 10,  left: 10,  bottom: 10,  right: 10)
        ));
        lblResultado.setOpaque(true);
        lblResultado.setBackground(Color.WHITE);
        gbc.gridy = 4; gbc.gridx = 0; gbc.gridwidth = 2;
        panelPrincipal.add(lblResultado, gbc);

        add(panelPrincipal);
    }

    public JTextField getTxtCelsius() { return txtCelsius; }  2 usages
    public JButton getBtnFahrenheit() { return btnFahrenheit; }  1 usage
    public JButton getBtnKelvin() { return btnKelvin; }  1 usage
```

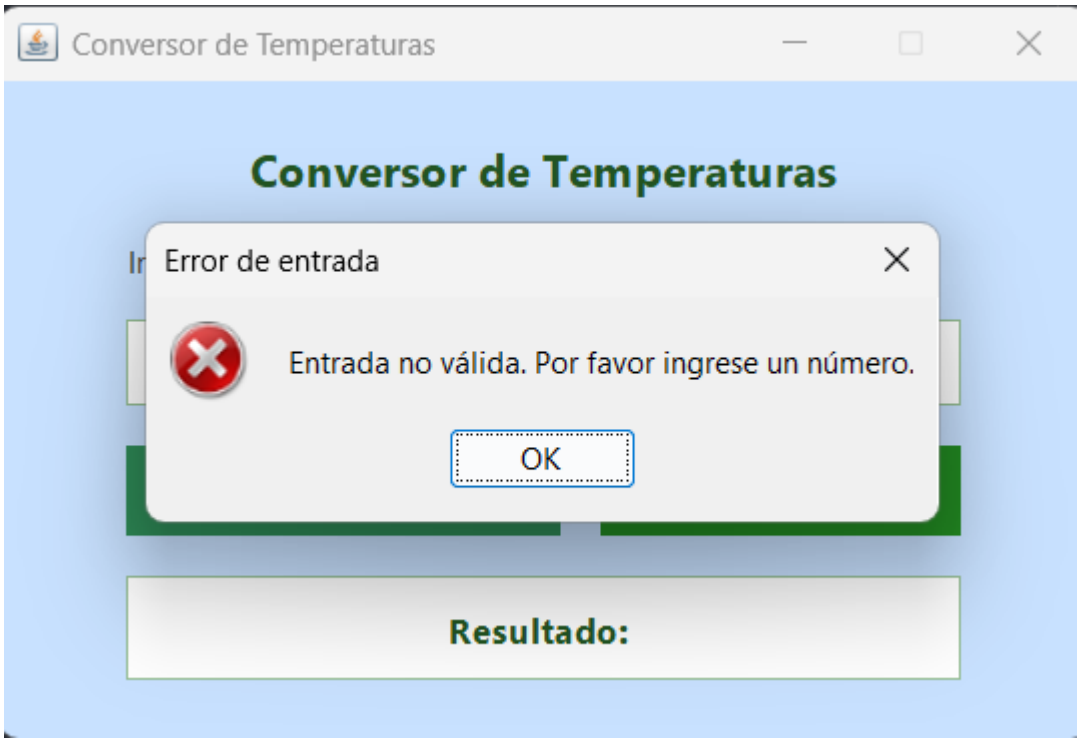
```java
    Edit | Explain | Test | Document | Fix
    public void setResultado(String texto, Color color) {  2 usages
        lblResultado.setText(texto);
        lblResultado.setForeground(color);
    }

    public void mostrarError(String mensaje, String titulo) {  1 usage
        JOptionPane.showMessageDialog( parentComponent: this, mensaje, titulo, JOptionPane.ERROR_MESSAGE);
    }

    public void mostrarAviso(String mensaje, String titulo) {  1 usage
        JOptionPane.showMessageDialog( parentComponent: this, mensaje, titulo, JOptionPane.WARNING_MESSAGE);
    }
}
```

Main.java

```java
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

Edit | Explain | Test | Document | Fix
public class Main {
    private ConversorView vista;  9 usages

    Edit | Explain | Test | Document | Fix
    public Main(ConversorView vista) {  1 usage
        this.vista = vista;
        inicializarEventos();
    }


    private double celsiusAFahrenheit(double celsius) {  1 usage
        return (celsius * 9.0 / 5.0) + 32;
    }

    private double celsiusAKelvin(double celsius) {  1 usage
        return celsius + 273.15;
    }
```

```java
private void inicializarEventos() {  1 usage
    vista.getBtnFahrenheit().addActionListener(new ActionListener() {
        Edit | Explain | Test | Document | Fix
        @Override
        public void actionPerformed(ActionEvent e) {
            convertir( tipo: "fahrenheit");
        }
    });

    vista.getBtnKelvin().addActionListener(new ActionListener() {
        Edit | Explain | Test | Document | Fix
        @Override
        public void actionPerformed(ActionEvent e) {
            convertir( tipo: "kelvin");
        }
    });
}

Edit | Explain | Test | Document | Fix
private void convertir(String tipo) {  2 usages
    String entrada = vista.getTxtCelsius().getText().trim();

    if (entrada.isEmpty()) {
        vista.mostrarAviso(
                mensaje: "Por favor ingrese un valor de temperatura.",
                titulo: "Campo vacío"
        );
        return;
    }
```

```java
Edit | Explain | Test | Document | Fix
public static void main(String[] args) {
    try {
        javax.swing.UIManager.setLookAndFeel(
                javax.swing.UIManager.getSystemLookAndFeelClassName()
        );
    } catch (Exception e) {
        // Continúa con look & feel por defecto
    }

    javax.swing.SwingUtilities.invokeLater(new Runnable() {
        Edit | Explain | Test | Document | Fix
        @Override
        public void run() {
            ConversorView vista = new ConversorView();
            new Main(vista);
            vista.setVisible(true);
        }
    });
}
}
```

Código en Ejecución

Ejemplo de Validación: